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GPS-Mamba: Graph permutation scanning state space model for multivariate time series forecasting

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ABSTRACT

Modeling complex correlations between variables in multivariate time series forecasting is an increasingly recognized challenge. Transformer-based models have shown leading performance, but encounter obstacles of weak inductive bias and quadratic computational complexity. Recently emerged State Space Models, such as Mamba, offer comparable performance to Transformer with higher efficiency. Nevertheless, existing methods still face three key challenges. First, most models lack comprehension of graph positional inductive bias. Secondly, the coexistence of valuable information and noise hinders correlation extraction. Furthermore, representative invariant patterns in multivariate time series are overlooked in dynamic modeling. To address these issues, we present a novel Mamba framework called GPS-Mamba to effectively model multivariate correlations. Specifically, we propose the Laplace Scanning algorithm that introduces graph structure information and multivariate correlations into permutation to enhance positional inductive bias. Based on the algorithm, we utilize Mamba-based encoder to model long-term stable correlations and short-term dynamic variation with devised retrieval mechanism, respectively. Finally, we combine the features obtained from two perspectives to achieve more precise prediction. Experimental results on seven real-world datasets demonstrate the superiority of GPS-Mamba over SOTA baselines with high interpretability. The codes are available at https://github.com/yzh8221/GPS-Mamba.

1. Introduction

Multivariate time series (MTS) consists of multiple series spanning over multiple time steps. Multivariate time series forecasting (MTSF) involves estimating future MTS based on past observations. It has wide applications in a variety of fields, including air quality prediction (Hettige et al., 2024), energy management (Zhou et al., 2021) and intelligent transportation systems (Shao et al., 2024).

MTSF methods typically involve modeling correlations on variable dimension (Qiu et al., 2025). It substantially affects prediction accuracy in data with complex multivariate correlations, especially those exhibiting spatial indistinguishability within the dynamic characteristics (Shao et al., 2024). As shown in Fig. 1(b), the correlation matrix constructed from traffic flow exhibits dynamic variations, with certain variables manifest similar patterns during morning peak periods but not necessarily at other times. Models without considering multivariate correlations modeling may degrade performance on such datasets (Shao et al., 2024).

With the rise of Transformer architecture (Liu et al., 2024a; Vaswani et al., 2017) and its success in diverse fields (Dosovitskiy, 2020), it was introduced into MTSF task and demonstrated great power benefiting from self-attention mechanism. Initially, temporal modeling methods like Autoformer (Wu et al., 2021) and FEDformer (Zhou et al., 2022) treat different variables at the same time step as tokens to implicitly model multivariate correlations, but potentially losing meaningful information. Subsequently, the channel-independent strategy has been adopted by methods such as DLinear (Zeng et al., 2023) and PatchTST (Nie et al., 2023) and demonstrated effectiveness on certain datasets. Nevertheless, such strategy discards potentially relevant inter-channel relationships. To this end, Crossformer (Zhang & Yan, 2023) and iTransformer (Liu et al., 2024c) endeavor to handle variable correlations through channel attention and has been verified to be effective. However, the standard global attention of Transformer calculates dependencies on each pair of variables and relies on positional encoding to capture structural information (Behrouz & Hashemi, 2024). Thus, Transformer-based models still face challenges with weak inductive bias and quadratic complexity, limiting their applicability in realistic scenarios (Fang et al., 2025).

Recently, State Space Models (SSMs), as an alternative of attention-based models have attracted increasing popularity. SSMs are principled sequential models describing the evolution of states over time (Gu et al., 2021b). Its prohibitive computation and memory requirements render

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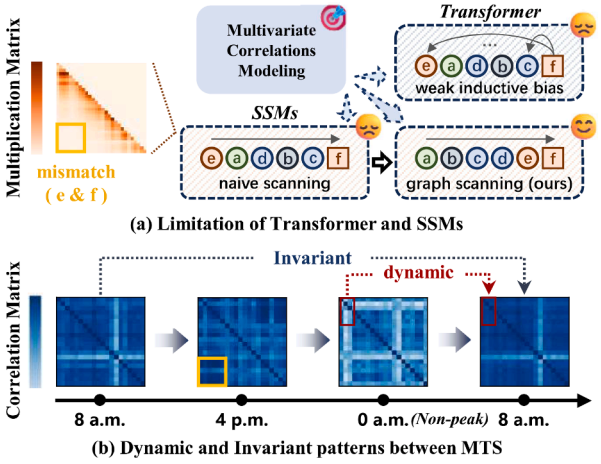


Fig. 1. Illustration of motivating examples. (a) In multivariate correlation modeling, Transformer exhibits weak inductive bias and is prone to introducing noise, while the naive scanning of SSMs struggles to accommodate non-sequential structure. (b) Invariant features within the dynamic variations of variable correlations.

it infeasible as a general sequence modeling solution. To address the limitations, structured SSMs such as S4 (Gu et al., 2021a) introduces the reparameterization technique to improve computational efficiency. However, they hardly achieve competitive performance to Transformer due to the input-dependent context compression (Liu et al., 2025). To overcome the restriction, Mamba (Gu & Dao, 2023) is proposed with selection mechanism that parameterizes input tokens to filter out irrelevant information. Motivated by its success in natural language processing, S-Mamba (Wang et al., 2025) and MambaMixer (Behrouz et al., 2024) further substantiate the capability of Mamba in modeling multivariate correlations via mitigating redundant information propagation across variables.

Despite the progress achieved, existing MSTF methods still encounter three main challenges:

1) Insufficient comprehension of positional inductive bias. Irregularly distributed sensors in real-world can always form graph or other implicit structure, which is non-negligible in MTSF. As shown in Fig. 1(a), Transformer allows interaction between arbitrary tokens regardless of their positions. Such weak inductive bias may lead to performance degradation. In contrast, the recurrent scanning paradigm of SSMs naturally considers tokens position (Behrouz & Hashemi, 2024). However, the sequential paradigm makes it challenging to directly apply SSMs for modeling non-causal graph structures and multivariate correlations. Consequently, rather than naively scanning the default sequence permutation, graph scanning incorporating structure information will better benefit downstream application tasks.

2) Coexistence of valuable information and noise between variables hinders correlation extraction. Global attention on all channels aims to extract information comprehensively, but may simultaneously introduce noise and reduce model's robustness (Qiu et al., 2025). On the other hand, selection mechanism in Mamba can filter out harmful noise. Meanwhile, as the SSMs paradigm of Mamba compresses information into finite latent states, crucial information may be lost during long scanning intervals as new tokens continuously influx into the finite space. As shown in Fig. 1(b), the modeling of a particular channel can be enhanced by exploiting information from other related channels (highlighted in yellow). However, the SSM multiplication matrix heatmap in Fig. 1(a) reveals that variables separated by long scanning intervals (e.g., e and f) are tend to be assigned low values. Intuitively, channels with stronger correlations should receive higher weights, and this mismatch hinders the retention of critical information during scanning.

3) Representative invariant patterns across dataset are overlooked in dynamic modeling. Dynamic correlation has been widely explored in numerous studies, which aim to reveal multivariate correlations of each time slice and dynamically propagate variable information over time (Fang et al., 2025). Most methods suffer from high computational complexity and erratic output fluctuations (Chen et al., 2024b). In fact, extensive dynamic multivariate correlations can be distilled into a few representative patterns, which contribute to the model's convergence and expressive capacity (Chen et al., 2024b). As illustrated in Fig. 1(b), the correlation matrices of MTS vary over time while similar patterns emerge at the same periods across different days.

To address the aforementioned issues, we present a novel Mamba framework called GPS-Mamba (**G**raph **P**ermutation **S**canning Selective State Space Model) for MTSF task. We first propose a Laplace Scanning algorithm to incorporate graph structure information (or multivariate correlations) into sequence permutation. Next, we utilize Mamba block with the proposed algorithm to encode multivariate correlations from different perspectives. Since the distance graph and long-term variable correlations represent the same underlying stable feature (Deng et al., 2021), we use the scanning mode derived from them for modeling long-term multivariate correlations. For short-term variation modeling, we construct a repository of representative samples from training set and employ a retrieval mechanism to model dynamic multivariate correlations. Finally, features from two perspectives are fused to enhance model's predictive capability. Notably, GPS-Mamba is interpretable due to the scanning modes informed by domain knowledge of irregularly distributed graphs and correlation matrices. To summarize, the main contributions of this paper are as follows:

- We present GPS-Mamba, a generic Mamba framework with retrieval mechanism tailored for dynamic and effective multivariate correlations modeling for MTSF task.
- We introduce Laplace Scanning, an algorithm that explicitly incorporates graph structure and correlation information into sequence modeling, a factor neglected by existing MTSF models. Moreover, it serves as a plug-and-play module that can seamlessly integrate with SSMs.
- We conduct experiments on seven publicly available benchmark datasets. Results demonstrate that GPS-Mamba performs well in terms of prediction precision with great interpretability.

2. Related work

In this section, we first review the pertinent work on multivariate correlations modeling and then introduce state space models with their corresponding scanning modes.

2.1. Multivariate correlations modeling

It is essential to consider multivariate correlations modeling in MTSF (Shao et al., 2024). Previous models (Wu et al., 2021; Zhou et al., 2021, 2022) opt to treat different variables at the same time step as tokens to implicitly model multivariate correlations, leading to the loss of meaningful information and suboptimality (Zhang & Yan, 2023). DLinear (Zeng et al., 2023), PatchTST (Nie et al., 2023) and TimeMachine (Ahamed & Cheng, 2024) demonstrate the effectiveness of channel independent strategy in prediction on certain datasets. However, the assumption of ignoring channel dependencies exacerbates performance degradation on datasets with strong correlations and spatial indistinguishability (Qiu et al., 2024; Shao et al., 2024). Subsequently, GNN-based approaches (Yi et al., 2024) attempt to model cross-variable correlations using graph structures, yet most are limited in capturing spatial dependencies across time steps and are susceptible to noise in multivariate correlations (Huang et al., 2023). Meanwhile, Crossformer

(Zhang & Yan, 2023), Leddam (Yu et al., 2024) and iTransformer (Liu et al., 2024c) embed variables into tokens with attention mechanism to model variate correlations and have achieved impressive results. Nevertheless, interaction effects among certain variables in MTS are relatively weak. Therefore, modeling interaction between arbitrary variables could introduce detrimental noise (Qiu et al., 2025). To focus solely on extracting dependencies beneficial to prediction, recent studies utilize patch-based methods (Fang et al., 2025) and cluster-based methods (Chen et al., 2024a) that group variables and model correlations within each group, which inevitably cause information loss due to dynamic multivariate correlations. To this end, DUET (Qiu et al., 2025) captures relationships within evolved cluster. Sumba (Chen et al., 2024b) extracts dynamic representations via structured patterns. Despite these efforts, the weak inductive bias of Transformer (Behrouz & Hashemi, 2024) and inflexible cluster-based methods remain challenges (Qiu et al., 2025), which inspires us to explore SSMs for MTSF.

2.2. State space models

State space models are a traditional mathematical framework depicting the system dynamic behavior (Qu et al., 2024) (Patro & Agneeswaran, 2024). The core idea of SSMs is embodying system's behavior through a collection of hidden variables (Hamilton, 1994) to effectively capture data dependencies. LSSL (Gu et al., 2021b) demonstrates the relationship between Recurrent and Convolutional models with Linear State-Space Layers. Later, several variants of SSMs are proposed to mitigate the issue of resource scarcity. RWKV (Peng et al., 2023) combines efficient parallelizable training of transformers with the efficient inference of RNNs. S4 (Gu et al., 2021a) uses reparameterization technique to improve the efficiency and scalability of SSMs. However, most of them miss the context due to linear time invariance mechanism (Hu et al., 2025). To improve context-aware capability, Mamba (Gu & Dao, 2023) introduces a selection mechanism to filter specific data by parameterizing input with efficient hardware-aware algorithm.

2.3. Scanning mode in SSMs

The unidirectional sequence scanning paradigm of SSMs hinders comprehensive modeling of non-sequential data (Behrouz & Hashemi, 2024). Numerous works have extended SSMs to different data modalities with specially designed scanning algorithms. In the domain of computer vision, models typically scan entire images using various types of curves in multiple directions (Hu et al., 2025; Qu et al., 2024). However, the irregular distribution of MTS is not suited for predefined scanning modes. For graph data, GMN (Behrouz & Hashemi, 2024) and Graph-Mamba (Wang et al., 2024a) formulate node prioritization and permutation strategies based on node degrees in graph. Nevertheless, they primarily focus on unweighted graphs and struggle to adapt to time series that lack an explicit graph structure. To better align Mamba with multivariate correlations modeling, SOR-Mamba incorporates regularization to minimize the discrepancy between embedding vectors generated from different channel orders. MambaTS explores various combinations of scan orders and evaluates their effectiveness during training, enabling adaptive adjustment of the optimal permutation during inference. However, it does not explicitly leverage available graph structure or multivariate correlations, and the exhaustive exploration of numerous combinations significantly increases computational overhead and introduces randomness. To this end, we propose the Laplace Scanning algorithm. It preserves adjacent attribute in graph structure or multivariate correlation space in sequence permutation, thereby efficiently propagating critical information during scanning. Furthermore, we utilize the proposed algorithm to encode both long-term correlations and short-term variations, facilitating a more comprehensive modeling of dynamic multivariate correlations.

3. Preliminaries

3.1. Problem definition

MTSF aims to predict future MTS value based on historical observations. Specifically, the historical input across N variables is denoted by $X = \{x_1^t, x_2^t, \dots, x_N^t\}_{t=1}^{L} \in \mathbb{R}^{L\times N}$, where $x_i^j \in \mathbb{R}$ represents the values of ith variable at time step j, L denotes the length of look-back window. The future data for next T time slices are defined as $Y = \{y_1^t, y_2^t, \dots, y_N^t\}_{t=L+1}^{L+T} \in \mathbb{R}^{T\times N}$. The forecasting model f_θ utilizes X to estimate the future values $\bar{Y}$.

3.2. State space models

SSMs are typically known as linear time-invariant systems, mapping input signals $x(t) \in \mathbb{R}^L$ to response sequence $y(t) \in \mathbb{R}^L$ through latent states $h(t) \in \mathbb{R}^{N\times L}$. Here, $t > 0$ represents the time index, and N indicates the dimension of the state variable. The dynamic system can be represented as an ordinary differential equation (ODE) (Gu et al., 2021b):

$$h'(t) = \mathbf{A}h(t) + \mathbf{B}x(t), \tag{1}$$

$$y(t) = \mathbf{C}h(t) + \mathbf{D}x(t), \tag{2}$$

where $h'(t)$ is the derivative of state $h(t)$, evolution parameter $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix, $\mathbf{B} \in \mathbb{R}^{N\times 1}, \mathbf{C} \in \mathbb{R}^{1\times N}$ are projection parameters, and coefficient $\mathbf{D} \in \mathbb{R}$ determines the direct effect of the input on the output. Given that the term $\mathbf{D}x$ can be viewed as a skip connection in deep learning, most SSMs exclude it by assuming $\mathbf{D} = 0$.

Due to the hardness of finding analytical solutions for solving above ODE and the challenge in integrating continuous-time SSMs with deep learning frameworks, Zero-Order Hold (ZOH) discretization (Gu & Dao, 2023) is applied to approximate the continuous-time SSMs into a discrete-time form. Specifically, given a timescale Δ representing the interval $[t-1, t]$, the function value is assumed to remain constant over Δ. The discretized SSMs (Gu et al., 2021a) can be represented as:

$$h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t, \tag{3}$$

$$y_t = \mathbf{C}h_t, \tag{4}$$

where $\bar{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\bar{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - \mathbf{I}) \cdot \Delta\mathbf{B}$.

By creating convolutional kernels $\bar{\mathbf{K}}$, the discrete-time SSMs are equivalent to convolution form (Gu et al., 2021a) that facilitates efficient computation:

$$\bar{\mathbf{K}} = (\bar{\mathbf{C}}\bar{\mathbf{B}}, \bar{\mathbf{C}}\bar{\mathbf{A}}\bar{\mathbf{B}}, \dots, \bar{\mathbf{C}}\bar{\mathbf{A}}^{L-1}\bar{\mathbf{B}}), \tag{5}$$

$$y = x * \bar{\mathbf{K}}, \tag{6}$$

Since time-independent parameters lack content-aware representation, Mamba introduces a selective mechanism that makes $\mathbf{B}, \mathbf{C}$, and Δ depend on the input x_t. However, input-dependent convolutional kernels results in the inability to perform parallel computations. To achieve efficient computation, Mamba further utilizes hardware-aware computation algorithm with linear complexity.

3.3. Formalization of scanning mode

The unidirectional sequence scanning paradigm of SSMs implies that data permutation could have an impact on model performance. Here, we formalize the process of scanning variables for multivariate correlations modeling as follows.

A dataset X with variable permutation $P = \{i_1, i_2, \dots, i_n\}$ can be represented as $X = \{x_{i_1}, x_{i_2}, \dots, x_{i_n}\}$. Without loss of generality, the permutation P is set to $\{1, 2, \dots, n\}$ initially. Applying a new permutation on X through certain algorithms can be represented as a bijective function π defined on P:

$$\pi : P \to \pi(P) = \{p_1, p_2, \dots, p_n\}, \tag{7}$$

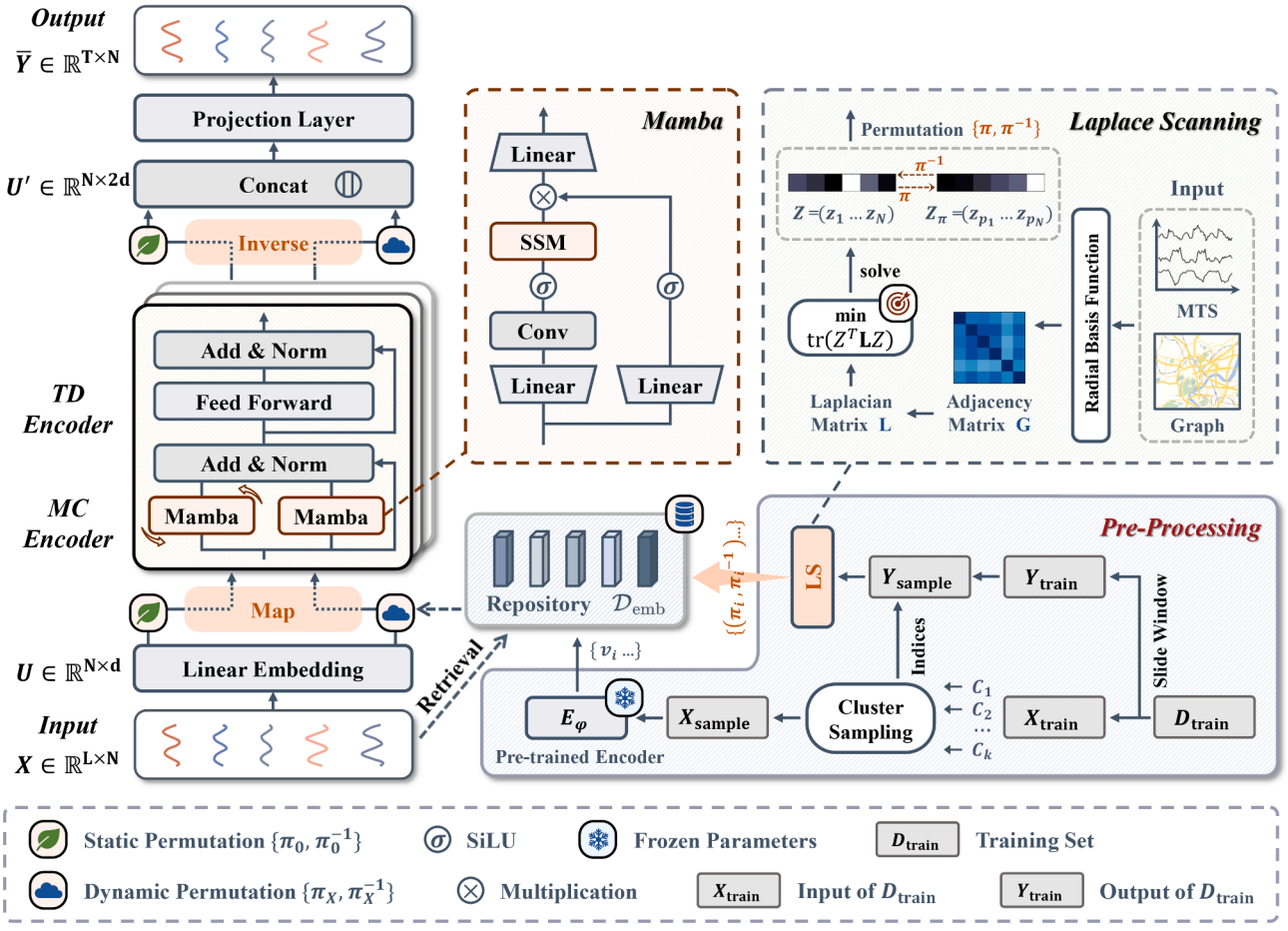


Fig. 2. Overview the architecture of GPS-Mamba. It consists of pre-processing and training phases. In pre-processing phase, the representative patterns and corresponding scanning modes are extracted from training set D_{train} and stored in the repository. In training phase, the Mamba-based encoder module models long-term multivariate correlations and short-term variations, respectively. The former leverages stable features derived from graph structure or long-term dependencies as the static permutation scanning mode, while the latter retrieves representative pattern from the devised repository as the dynamic scanning mode.

$$\pi^{-1} : \pi(P) \to P = \{1, 2, \dots, n\}. \tag{8}$$

SSMs scan multivariate along the new specified permutation $\pi(P)$ can be formulated as $f(X, \overrightarrow{\pi(P)})$, where function f corresponds to the SSMs, $\overrightarrow{\pi(P)}$ represents the forward order with respect to the given variable permutation $\pi(P)$. Correspondingly, $\overleftarrow{\pi(P)}$ denotes the inverse order of $\pi(P)$.

4. Methodology

In this section, we present the overall architecture of our GPS-Mamba depicted in Fig. 2. It consists of pre-processing and training phases. In the pre-processing phase, we extract representative patterns from the training set, and store corresponding scanning modes obtained by the proposed Laplace Scanning algorithm in a repository. In the training phase, we integrate Mamba block with encoder layer to model multivariate correlations and temporal dependencies from two views. During long-term multivariate correlations modeling, the static scanning mode of Mamba is derived from the stable feature of MTS. During short-term variation modeling, we employ a retrieval mechanism to dynamically match representative scanning modes from the retrieval repository. Finally, we fuse features from two parts and employ a projection as decoder to predict future MTS. The main notations used in this article are listed in Table 1. Detailed description of key components of the proposed methods are as follows.

4.1. Repository construction and retrieval

4.1.1. Repository construction

In pre-processing stage shown in Fig. 2, we construct a representative sample repository across training set for dynamic correlations modeling.

Initially, the training set $D_{\text{train}} \in \mathbb{R}^{H\times N}$ is divided into historical MTS $X_{\text{train}} \in \mathbb{R}^{(H-2K+1)\times K\times N}$ and corresponding future MTS $Y_{\text{train}} \in \mathbb{R}^{(H-2K+1)\times K\times N}$ by slide window, where K denotes the length of slide window. As mentioned in the introduction, the patterns of data will change over time, but similar characteristics will emerge within the same period on different days. Therefore, we initialize k subsets in X_{train} based on time intervals within a day:

$$\mathcal{D} := \{c_1, c_2, \dots, c_k \mid \forall c_i \subseteq X_{\text{train}}\} \tag{9}$$

where c_i represents instances that occur within the period i of a day, k is a hyperparameter to regulates the duration of time period. For example, when the time interval is one hour, k is set to 24 and c_8 refers to the aggregated time series from 7 : 00 am to 8 : 00 am.

Within period instances in c_i, several distinct patterns can be identified by cluster algorithm. To avoid the selection of a predefined number of clusters leading to the omission of certain category patterns, we employ DBSCAN on the t-SNE results of instances in c_i and finally obtain a total of m representative clusters:

$$\mathcal{D}' = \cup_{i=1}^{k} \text{DBSCAN}(c_i) = \{c'_1, c'_2, \dots, c'_i, \dots, c'_m\} \quad , \tag{10}$$

where c'_i denotes the ith cluster and m is the number of clusters adaptively estimated DBSCAN.

Table 1
Main notations used in article.

Notations	Descriptions
X, $\bar{Y}$	historical input of model, output of the model
X_i, x_i^j	time series of ith variable, i^{th} variable at time step j
X_{train}, Y_{train}	input training set, output training set
X_{sample}, Y_{sample}	input and output of representative patterns in X_{train}
$\mathbf{G}$, g_{ij}	adjacency matrix of variables, edge weight of $\mathbf{G}$
d_{ij}	distance between variable i and j
θ, ϵ, σ	radial basis factor, threshold factor, scaling factor
Z, z_i	variables mapping vector, mapping value of ith variable
$\mathbf{L}$, $\mathbf{D}$	laplacian matrix of $\mathbf{G}$, degree matrix of $\mathbf{G}$
π, π^{-1}	permutation reorder function and inverse function
π_0, π_0^{-1}	permutation function in long-term correlation modeling
π_X, π_X^{-1}	permutation function in short-term variation modeling
Z_π	vector of applying function π to vector Z
$\mathcal{D}$	collection of subsets dividing X_{train} based on period
$\mathcal{D}'$	collection of subsets containing multiple clusters
$\mathcal{D}_{\text{emb}}$	repository of embedding, index and scanning function
c_i, k	instance of ith time period, number of time periods
c_i', m, n_s	i^{th} cluster in $\mathcal{D}'$, number of cluster, number of sample
E_ϕ, v_i, v	pre-trained encoder, embedding of ith sample and X
U^l, f	output of lth layer encoder, Mamba module
P, $\pi(P)$	original permutation, permutation applying function π
$\overrightarrow{\pi(P)}$, $\overleftarrow{\pi(P)}$	forward order and inverse order of $\pi(P)$
l_1, l_2	number of encoder layers in long-term correlation modeling and short-term variation modeling phases

We apply cluster sampling (Sedgwick, 2014a) on $\mathcal{D}'$ to obtain n_s representative input samples X_{sample} and corresponding Y_{sample} sharing same indices. X_{sample} is employed for dynamically matching model's inputs. To effectively represent X_{sample}, we utilized a pre-trained encoder E_φ(e.g., S-Mamba Wang et al., 2025) with its parameter set φ frozen. The embedding of X_{sample} is subsequently stored in repository $\mathcal{D}_{\text{emb}}$. As for Y_{sample}, it represents foresee genuinely correlated variables which are more conducive to precise modeling compared to X_{sample}. For instance, if the time period of input is not during traffic peak hours, while the predicted time period is a rush-hour, pre-judging such a situation will be conducive to better capturing the actual correlation between variables. Thus, we obtain the permutation function set of Y_{sample} by applying Laplace Scanning algorithm (as described in next Section 4.2) and store it in $\mathcal{D}_{\text{emb}}$. The repository $\mathcal{D}_{\text{emb}}$ can be expressed as:

$$\mathcal{D}_{\text{emb}} = \{\{i, v_i, \pi_i, \pi_i^{-1}\} \mid v_i = E_\varphi(X^i_{\text{sample}})\} \tag{11}$$

where X^i_{sample} denotes the ith representative sample in X_{sample}, v_i represents the embedding of X^i_{sample}, π_i and π_i^{-1} are corresponding permutation functions.

4.1.2. Retrieval mechanism

Different scanning modes of SSMs could implicitly model dynamic multivariate correlations. However, totally different scanning mode for every varying input may lead to instability and suboptimality. Since similar inputs often have similar patterns, we can retrieve representative scanning modes from repository $\mathcal{D}_{\text{emb}}$ based on the input.

We quantify the degree of match by capturing the Euclidean norm between the input embedding v and the representative sample embeddings in $\mathcal{D}_{\text{emb}}$, where $v = E_\varphi(X)$ represents the embedding of input X. Then, we can retrieve the permutation mapping functions corresponding to the smallest difference:

$$\text{index}(v) = \underset{v_i \in \mathcal{D}_{\text{emb}}}{\arg\min} \|v - v_i\|_2 \tag{12}$$

$$(\pi_X, \pi_X^{-1}) = (\pi_{\text{index}(v)}, \pi^{-1}_{\text{index}(v)}) \tag{13}$$

where index$(\cdot)$ represents the retrieved index given $\mathcal{D}_{\text{emb}}$, (π_X, π_X^{-1}) are the retrieved permutations. We summarize the procedure in Algorithm 1.

Algorithm 1 Repository construction and retrieval.

Input: Multivariate time series X, training set $\mathcal{D}_{\text{train}}$, pre-trained encoder E_φ, hyperparameter k and n_s
Output: Permutation functions (π_X, π_X^{-1})
1: Divide $\mathcal{D}_{\text{train}}$ into X_{train} and Y_{train} by slide window
2: Initialize k subsets in $\mathcal{D}$ based on X_{train}
3: Cluster instances in $\mathcal{D}$ to construct $\mathcal{D}'$ via Eq. (10)
4: Obtain X_{sample} and Y_{sample} from $\mathcal{D}'$ by cluster sampling
5: Obtain function set $\{\ldots, (\pi_i, \pi_i^{-1}), \ldots\}$ of Y_{sample}
6: Obtain $\mathcal{D}_{\text{emb}} = \cup_{i=1}^{n_s}\{i, v_i, \pi_i, \pi_i^{-1} | v_i = E_\varphi(X^i_{\text{sample}})\}$
7: Encode input MTS to embedding $v = E_\varphi(X)$
8: Obtain index(v) of the embedding based on Eq. (12)
9: **return** Functions $(\pi_X, \pi_X^{-1}) = (\pi_{\text{index}(v)}, \pi^{-1}_{\text{index}(v)})$

It is noteworthy that, during both training and inference, GPS-Mamba retrieves matching permutations from $\mathcal{D}_{\text{emb}}$ based on input. While this process may indirectly allow future information to influence the permutation selection for the training data, it can be considered a reasonable strategy to fully leverage the available training set (Liu et al., 2024b). During inference, test data retrieves only permutations from the repository $\mathcal{D}_{\text{emb}}$ constructed from training set, thus preventing leakage of future information from test set.

4.2. Laplace Scanning (LS) algorithm

Naive scanning mode of SSMs cannot properly reflect the graph structure or multivariate correlations within the sequence permutation. Intuitively, a rational strategy should preserve adjacent attribute in graph structure or multivariate correlation space during permutation; in other words, it should place highly correlated variables in proximity during scanning to facilitate critical information propagation, as the weights in the multiplication matrix decay exponentially with increasing scanning intervals. Therefore, the main goal of the proposed Laplace Scanning algorithm is to design the sequence permutation function based on multivariate correlations. Fig. 2 shows the process of the Laplace Scanning algorithm.

Firstly, we unify the graph or MTS of the algorithm input into an adjacency matrix $\mathbf{G}$. For graph structure with distance information between variables, the element g_{ij} of adjacency matrix $\mathbf{G}$ can be constructed by Gaussian radial basis function (Deng et al., 2021):

$$g_{ij} = \begin{cases} \frac{\exp\left(-\left\|d_{ij}\right\|_2\right)}{\theta}, & \text{if } d_{ij} < \epsilon \\ 0, & \text{otherwise} \end{cases}, \tag{14}$$

where d_{ij} denotes the distance between node i and node j, θ is a hyperparameter to control the distribution, ϵ is a predefined threshold to control the sparsity of $\mathbf{G}$.

As for MTS without graph distance information, a similarity-based graph can be constructed and the adjacency matrix $\mathbf{G}$ can be obtained by radial basis function kernel (Chen et al., 2024a):

$$g_{ij} = \exp\left(\frac{-||X_i - X_j||^2}{2\sigma^2}\right). \tag{15}$$

X_i represents time series of variable i, σ is a scaling factor.

The core idea is to keep physically nearby or similar variables adjacent during scanning, while separating unrelated variables apart. Since sequence permutation is one-dimensional, we assume that each variable i can be represented by a real value $z_i \in \mathbb{R}$, and g_{ij} represents the closeness between variables i and j. Then the difference between z_i and z_j should approximate g_{ij}. The goal is to minimize the overall disparity among all variables, which leads to solving the objective function:

$$\min \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\left\|z_i - z_j\right\|^2 g_{ij}. \tag{16}$$

This objective function can be expanded to its simplified form:

$$\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\left\|z_i - z_j\right\|^2 g_{ij} = \mathrm{tr}\left(Z^\top \mathbf{L} Z\right), \tag{17}$$

where $Z = (z_1, z_2, \ldots, z_N)^\top \in \mathbb{R}^N$ represents the mapping vector of variables, $\mathbf{L}$ denotes the Laplacian matrix of $\mathbf{G}$, $\mathrm{tr}(\cdot)$ represents the trace of matrix. To ensure that the mapped value is distributed throughout mapping space rather than converging, additional constraint is introduced:

$$Z^\top \mathbf{D} Z = \mathbf{I}, \tag{18}$$

where $\mathbf{D}$ is the degree matrix of $\mathbf{G}$. Then we can use Lagrange multipliers (Rockafellar, 1993) to solve the objective function:

$$\begin{aligned} &f(Z) = \mathrm{tr}\left(Z^\top \mathbf{L} Z\right) + \mathrm{tr}\left[\mathbf{\Lambda}\left(Z^\top \mathbf{D} Z - \mathbf{I}\right)\right], \frac{\partial f(Z)}{\partial Z} = 0 \\ &\Rightarrow \mathbf{L} Z = -\mathbf{D} Z \mathbf{\Lambda} \end{aligned}, \tag{19}$$

where $\mathbf{\Lambda}$ is a diagonal matrix with $\mathbf{\Lambda}_{ii} = \lambda_i$. Accordingly, it can be rewritten in the standard form:

$$\mathbf{L} z_i = \lambda_i \mathbf{D} z_i. \tag{20}$$

The optimization function is equivalent to the generalized eigenvalue problem $\mathbf{L}z = \lambda \mathbf{D}z$. Ghojogh et al. (2019) The desired Z is the eigenvector corresponding to the smallest non-zero eigenvalue.

By sorting the elements of vector Z, the ordered vector Z_π is generated. Since the mapped values of correlated variables are numerically close, their corresponding permutation in Z_π is also adjacent, thus yielding the desired scanning permutation. The function mapping original variables order in Z to objective permutation in Z_π is noted by π, and π^{-1} denotes the corresponding inverse function.

4.3. Encoder layer

In this section, we present the structure of the encoder layer in GPS-Mamba. It takes $U \in \mathbb{R}^{N \times d}$ obtained from the embedding layer as input, and comprises Multivariate Correlation (MC) encoder and Temporal Dependencies (TD) encoder as shown in Fig. 2. The encoder modules for long-term correlation modeling and short-term variation modeling are almost identical, with the exception of the number of encoder layers. Moreover, the scanning modes in the MC encoder for the Mamba block are distinct. The static scanning mode π_0 in long-term correlation modeling is derived from either graph or long-term correlation using the proposed Laplace Scanning algorithm. In contrast, the scanning mode π_X in short-term variation modeling is dynamically retrieved from the repository D_{emb} based on the input. To simplify the representation and avoid confusion, we will use π to represent the scanning mode π_0 and π_X in the following encoder module illustrations.

4.3.1. Multivariate correlation encoder

Within this layer, the primary objective is to extract multivariate correlations by learning the mutual information, thereby enabling a more robust representation of variates. Here, we attempt to utilize the selective mechanism of Mamba block f to model the correlations and filter out irrelevant noise. The input U^{l-1} to Selective SSM f at layer l can be formulated as Behrouz and Hashemi (2024):

$$\begin{aligned} &\Phi = \sigma(\mathrm{Conv}(W_1\ \mathrm{LayerNorm}(U^{l-1}))) \\ &\mathbf{B} = W_{\mathbf{B}}\Phi, \quad \mathbf{C} = W_{\mathbf{C}}\Phi, \quad \Delta = \mathrm{Softplus}(W_\Delta \Phi), \\ &\overline{\mathbf{A}} = \mathrm{Discrete}_{\mathbf{A}}(\mathbf{A}, \Delta), \quad \overline{\mathbf{B}} = \mathrm{Discrete}_{\mathbf{B}}(\mathbf{B}, \Delta), \\ &y = \mathrm{SSM}_{\overline{\mathbf{A}},\overline{\mathbf{B}},\mathbf{C}}(\Phi), \\ &U^l = W_2(y \odot \sigma(W_3\ \mathrm{LayerNorm}(\Phi))), \end{aligned} \tag{21}$$

where $W_1, W_{\mathbf{B}}, W_{\mathbf{C}}, W_\Delta, W_2, W_3$ are learnable parameters, $\sigma(\cdot)$ is nonlinear activation function SiLU, $\mathrm{SSM}(\cdot)$ and $\mathrm{Discrete}(\cdot)$ are the state space model and ZOH discretization discussed in Section 3.2.

However, the recurrent nature of Mamba limits it to incorporating only the information from preceding tokens, which does not align with the non-causal recursion pattern between variables. To capture the global context and overall multivariate correlations, we incorporate an additional reverse Mamba module to form a bidirectional Mamba layer to acquire comprehensive information:

$$U^l = f(U^{l-1}, \overrightarrow{\pi(P)}) + f(U^{l-1}, \overleftarrow{\pi(P)}) \tag{22}$$

where $\overrightarrow{\pi(P)}$ and $\overleftarrow{\pi(P)}$ represent the forward order and inverse order of permutation $\pi(P)$ during scanning, f is the Mamba module. We connect with another residual network to form the output of this layer $U^l = U^l + U^{l-1}$.

4.3.2. Temporal dependencies encoder

Since simple models could model complex temporal dependencies (Zeng et al., 2023), we further process the output of the MC Encoder using linear-based network. Firstly, a normalization layer is employed to normalize all variate tokens to a Gaussian distribution, which diminishes the discrepancies caused by inconsistent measurements. It has been proven to facilitate the convergence and training stability of model (Liu et al., 2024c). Then, a feed-forward network (FFN) implicitly encodes temporal dependencies through dense non-linear connections. Finally, another normalization layer is applied to generate stationary representations.

Following the encoder with l_1 and l_2 layers for long-term correlation and short-term variation modeling, respectively, the resulting outputs $U_1^{l_1}$ and $U_2^{l_2}$ are derived.

4.4. Projection decoder

Based on the outputs $U_1^{l_1} \in \mathbb{R}^{N \times d}$ and $U_2^{l_2} \in \mathbb{R}^{N \times d}$ from the multivariate interaction and temporal dependencies modeling, we standardize their variable permutations to the original order using the inverse function:

$$U_1^{l_1} = \pi_0^{-1}(U_1^{l_1}), \quad U_2^{l_2} = \pi_X^{-1}(U_2^{l_2}) \tag{23}$$

The results are concatenated to obtain $U' \in \mathbb{R}^{N \times 2d}$, thus fusing the two view-specific representations. Subsequently, we exploit the direct multi-step forecasting for the decoder to predict the multi-step MTS at once (Zeng et al., 2023). We take Linear Projector as decoder that maps the time dimension from input length $2d$ to the predict sequence length T, followed by transposition to yield the final output:

$$\{\hat{y}_1^t, \ldots, \hat{y}_N^t\}_{t=L+1}^{L+T} = (\mathrm{Projector}(U_1^{l_1} \| U_2^{l_2}))^\top \tag{24}$$

5. Experiments

5.1. Experimental setup

5.1.1. Datasets

We verify the performance of GPS-Mamba on eight real-world public datasets ranging from various domains as listed in Table 2. They are widely evaluated in MTSF task, including Weather, ECL, Traffic used by Wu et al. (2021), Solar provided in Lai et al. (2018), SD partitioned from Liu et al. (2023), PEMS03, PEMS08, NN5 evaluated in Liu et al. (2024c), Qiu et al. (2024). Information about the datasets is as follows.

- **Weather:** It records every 10 mins for the whole year 2020 containing 21 meteorological indicators, such as humidity and air temperature.
- **ECL:** It collects the hourly electricity consumption of 321 residents from 2012 to 2014.
- **Traffic:** It collects the hourly occupancy rate of different lanes on San Francisco freeway from 2015 to 2016.
- **Solar:** It comprises the solar power production records every 10 mins from 137 plants in Alabama in 2006.
- **SD:** It records every 15 mins of traffic flow from 716 sensors in San Diego county in 2019.
- **PEMS03:** It is recorded by CalTrans PeMS using 358 detectors, which covers the traffic flow from September to December in 2018.
- **PEMS08:** It is recorded by CalTrans PeMS using 170 detectors on 8 roads, which covers the traffic flow from July to August in 2016.

Table 2
The statistics of benchmark datasets.

Datasets	Variants	Timesteps	Granularity	Time Interval
Weather	21	52,696	10 mins	2020/1/1 - 2021/1/1
ECL	321	26,304	1 h	2012/1/1 - 2015/1/1
Traffic	862	17,544	1 h	2015/1/1 - 2017/1/1
Solar	137	52,560	10 mins	2006/1/1 - 2007/1/1
SD	716	35,040	15 mins	2019/1/1 - 2020/1/1
PEMS03	358	26,208	5 mins	2018/9/1 - 2018/12/1
PEMS08	170	17,856	5 mins	2016/7/1 - 2016/9/1
NN5	111	791	1 day	1996/3/18 - 1998/5/18

Table 3
The correlation value of datasets.

Weather	ECL	Traffic	Solar	SD	PEMS03	PEMS08	NN5
1.59	1.80	1.96	2.00	1.96	2.00	1.98	1.63

- **NN5:** It is originated from the observation of daily withdrawals at 111 randomly selected different cash machines at different locations within England.

Additionally, we measure the correlation value (Lubba et al., 2019; Qiu et al., 2024) for each dataset. Datasets with high correlation values indicate greater possibility of sharing common trends or patterns between variables involved. As listed in Table 3, variables in the vast majority of datasets incorporate strong and complex multivariate correlations (Shao et al., 2024), which aligns with the evaluation emphasized by multivariate correlations modeling.

5.1.2. Baselines

To evaluate the effectiveness of GPS-Mamba, we conducted a comparison with the following state-of-the-art baseline models, categorized by methodology. (1) Mamba-based model: S-Mamba (Wang et al., 2025), KARMA (Ye et al., 2025), SDE-Mamba (Weng et al., 2025); (2) Transformer-based models: iTransformer (Liu et al., 2024c), Crossformer (Zhang & Yan, 2023), PatchTST (Nie et al., 2023); (3) Linear-based models: TimesMixer (Wang et al., 2024b), DLinear (Zeng et al., 2023); (4) GNN-based model: FourierGNN (Yi et al., 2024), CrossGNN (Huang et al., 2023); (5) Retrieval-based model: RAFT (Tire et al., 2024).

5.1.3. Evaluation metrics

We utilize two commonly adopted numerical metrics for all experiments to fairly compare models performance, including Mean Squared Error (MSE) and Mean Absolute Error (MAE) on each prediction window. The indicator formula is as follows:

$$\mathrm{MSE} = \frac{1}{T}\frac{1}{N}\sum_{i=1}^{T}\sum_{j=1}^{N}(y_j^i - \hat{y}_j^i)^2 \tag{25}$$

$$\mathrm{MAE} = \frac{1}{T}\frac{1}{N}\sum_{i=1}^{T}\sum_{j=1}^{N}|y_j^i - \hat{y}_j^i| \tag{26}$$

where y_j^i is the observed ground-truth value of jth variable at the ith time step, while $\hat{y}_j^i$ represents the corresponding model's predicted value.

5.1.4. Settings

We design experiments from the following aspects:

(1) To be consistent with the most modern methods, we split the datasets into training, validation and test set by the ratio of $6:2:2$ for SD and PEMS, and $7:1:2$ for Weather, ECL, Traffic, Solar and NN5. We consider look-back window $L = 36$ for NN5 and $L = 96$ for all other experiments. The length of prediction horizons $T \in \{12, 24, 48, 96\}$ for SD and PEMS, $T \in \{24, 36, 48, 60\}$ for NN5, and $T \in \{96, 192, 336, 720\}$ for Weather, ECL, Traffic and Solar.

(2) The training process was conducted for 10 epochs by default. To prevent overfitting, we employ early stopping with a counter that halts training if no significant improvement in loss on the validation set is observed for 3 consecutive epochs. We set hyper-parameter $\theta = 1$, radial basis factor $\epsilon = 0$, sample size $n_s = 50$. Scaling factor σ is set to standard deviation based on different datasets. The encoder layers l_1 and l_2 are selected from $\{2, 3, 4\}$, the dimension of hidden state in SSMs is chosen from $\{2, 8, 16, 32\}$.

(3) We rerun all the baselines using the source codes publicly released by the authors. For fair evaluation, we follow the default recommended settings of baselines in original papers and tune the key hyper-parameters when the settings are not specified.

(4) All experiments are implemented in Python with PyTorch 2.0.1 and conducted on a machine with the NVIDIA V100 GPU and 32GB memory using the Adam optimizer.

5.2. Overall performance

Table 4 showcases the MAE and MSE score across all methods on seven datasets with different horizon. The best scores are in bold and the second best are underlined. Based on the results, we have the following observations.

5.2.1. Effective multivariate correlations modeling

Compared to patch strategy and implicitly modeling dependency by embedding methods, introducing consideration for precise multivariate correlations modeling among variables can significantly improve the performance of MTSF on most datasets. The phenomenon highlights the necessity of leveraging precise multivariate correlations modeling in datasets with complex variable correlations. TimeMixer achieves optimum performance on specific settings. It may be attributed to the relatively small number of variables and insignificant correlations in Weather and Solar datasets. Therefore, modeling temporal dependencies is more conducive for prediction in these cases.

5.2.2. Superiority of selective SSM

Among the baselines concerned, SSMs with selection mechanism outperform Transformer-based models with attention mechanism. Especially for datasets characterized by high-dimensional variable space and complex multivariate correlations, such as SD and PEMS, Mamba-based models exhibit more pronounced improvements (over 15% MSE and 10% MAE improvement). The underlying cause is that dense connection modeling of global attention introduces noise from uncorrelated dimensions, leading to a decline in model performance. Besides, GNN-based methods may exhibit limitations in effectively capturing complex and dynamically evolving multivariate correlations. In contrast, selection mechanism enables model to parameterize the weight matrices based on the model input, thereby filtering out irrelevant information and selectively propagating relevant information during long-range modeling.

5.2.3. Overall best performance

Compared to S-Mamba, GPS-Mamba demonstrates systematic performance enhancements attributed to proposed scanning algorithm and explicit modeling of temporal periodicity dynamics. Remarkably, it establishes state-of-the-art performance or near-optimal results and achieves 0.22%-15.38% MSE improvement, and 0.29%-8.11% MAE improvement over all baselines on most of the datasets. As shown in Table 4, GPS-Mamba obtains 56 first-place and 17 second-place rankings in total 80 settings.

5.3. Ablation study

GPS-Mamba has three key components: long-term stable multivariate correlations (LTC) modeling, short-term dynamic variation (STV) modeling, and Laplace Scanning (LS) algorithm. To verify the effectiveness of each component, ablation experiments are conducted from three perspectives:

Table 4
Performance results of baselines and the proposed GPS-Mamba across datasets.

Method		GPS-Mamba (Ours)		S-Mamba (2025)		KARMA (2025)		SDE-Mamba (2025)		RAFT (2025)		iTransformer (2024)		Crossformer (2023)		PatchTST (2023)		TimeMixer (2024)		DLinear (2023)		FourierGNN (2023)		CrossGNN (2023)	
Metric		MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
Weather	96	0.165	**0.209**	0.165	0.210	0.168	0.210	0.165	0.213	0.191	0.241	0.176	0.216	0.183	0.256	0.174	0.215	**0.162**	0.210	0.196	0.255	0.174	0.248	0.180	0.230
	192	0.212	**0.251**	0.216	0.255	0.226	0.260	0.214	0.256	0.241	0.284	0.225	0.257	0.226	0.299	0.222	0.257	**0.209**	0.253	0.239	0.297	0.217	0.286	0.236	0.275
	336	0.270	**0.292**	0.274	0.297	0.280	0.299	0.273	0.298	0.296	0.326	0.281	0.299	0.281	0.344	0.278	0.297	**0.268**	0.295	0.283	0.332	**0.268**	0.328	0.289	0.313
	720	0.349	**0.345**	0.354	0.349	0.358	0.353	0.353	0.351	0.372	0.378	0.358	0.350	0.371	0.414	0.354	0.347	0.346	0.346	0.348	0.384	**0.344**	0.384	0.362	0.365
	Avg	0.249	**0.274**	0.252	0.278	0.258	0.281	0.251	0.280	0.275	0.307	0.260	0.281	0.265	0.328	0.257	0.279	**0.246**	0.276	0.267	0.317	0.251	0.312	0.267	0.296
ECL	96	0.140	**0.235**	**0.138**	0.236	0.143	0.237	0.147	0.244	0.166	0.268	0.148	0.239	0.153	0.254	0.180	0.273	0.157	0.249	0.195	0.277	0.211	0.306	0.173	0.275
	192	**0.160**	0.256	0.162	0.259	0.162	**0.253**	0.163	0.259	0.169	0.269	0.167	0.258	0.174	0.274	0.187	0.280	0.170	0.260	0.194	0.280	0.214	0.311	0.195	0.288
	336	0.184	0.277	0.180	0.277	**0.176**	**0.272**	0.177	0.275	0.180	0.280	0.179	**0.272**	0.222	0.314	0.204	0.296	0.187	0.277	0.207	0.296	0.228	0.326	0.206	0.300
	720	**0.204**	**0.296**	0.206	0.301	0.215	0.301	**0.204**	0.298	0.211	0.304	0.208	0.297	0.243	0.329	0.246	0.328	0.228	0.313	0.243	0.328	0.261	0.353	0.231	0.335
	Avg	**0.172**	**0.266**	**0.172**	0.268	0.174	**0.266**	0.173	0.269	0.182	0.280	0.176	0.267	0.198	0.293	0.204	0.294	0.186	0.275	0.210	0.295	0.229	0.324	0.201	0.300
Traffic	96	**0.379**	**0.260**	0.383	0.262	0.410	0.267	0.392	0.262	0.520	0.341	0.393	0.268	0.519	0.275	0.459	0.298	0.477	0.301	0.650	0.397	0.544	0.347	0.570	0.310
	192	**0.389**	**0.264**	0.394	0.267	0.448	0.278	0.415	0.272	0.498	0.322	0.413	0.277	0.550	0.288	0.469	0.301	0.490	0.308	0.600	0.372	0.538	0.343	0.577	0.321
	336	0.421	0.283	**0.420**	**0.278**	0.466	0.289	0.430	**0.278**	0.505	0.324	0.424	0.283	0.565	0.298	0.483	0.307	0.503	0.311	0.606	0.374	0.555	0.348	0.588	0.324
	720	**0.457**	0.300	0.466	**0.298**	0.510	0.308	0.463	**0.298**	0.541	0.353	0.458	0.300	0.597	0.334	0.517	0.326	0.525	0.318	0.646	0.396	0.596	0.364	0.597	0.337
	Avg	**0.412**	0.277	0.416	**0.276**	0.459	0.286	0.425	0.278	0.516	0.335	0.422	0.282	0.558	0.299	0.482	0.308	0.499	0.310	0.626	0.385	0.558	0.351	0.583	0.323
Solar	96	0.199	0.238	0.207	0.245	0.201	0.236	**0.188**	**0.219**	0.226	0.287	0.204	0.236	0.225	0.295	0.216	0.273	0.201	0.249	0.285	0.372	0.201	0.264	0.221	0.291
	192	0.228	**0.253**	0.241	0.273	0.230	0.256	0.233	**0.253**	0.247	0.311	0.239	0.263	0.244	0.291	0.249	0.292	0.231	0.271	0.315	0.397	**0.226**	0.286	0.236	0.300
	336	**0.247**	0.280	0.260	0.289	0.250	0.280	0.253	**0.270**	0.266	0.337	0.252	0.276	0.317	0.328	0.271	0.305	0.248	0.284	0.348	0.411	**0.247**	0.293	0.251	0.329
	720	0.251	0.277	0.262	0.291	0.253	0.281	**0.247**	**0.274**	0.261	0.332	0.251	0.277	0.325	0.374	0.262	0.295	0.255	0.289	0.351	0.409	0.248	0.294	0.253	0.315
	Avg	0.231	0.262	0.243	0.275	0.234	0.263	**0.230**	**0.254**	0.250	0.317	0.237	0.263	0.278	0.322	0.250	0.291	0.234	0.273	0.325	0.397	0.231	0.284	0.240	0.309
SD	12	**0.044**	**0.129**	0.048	0.137	0.066	0.152	0.057	0.144	0.081	0.182	0.052	0.137	0.055	0.146	0.076	0.177	0.051	0.139	0.125	0.216	0.074	0.177	0.056	0.142
	24	**0.060**	**0.146**	0.078	0.168	0.083	0.170	0.087	0.171	0.114	0.218	0.077	0.159	0.093	0.183	0.116	0.207	0.069	0.160	0.194	0.265	0.106	0.209	0.083	0.167
	48	**0.082**	**0.164**	0.104	0.184	0.112	0.181	0.110	0.190	0.146	0.246	0.103	0.174	0.150	0.235	0.161	0.234	0.094	0.181	0.246	0.298	0.144	0.239	0.108	0.182
	96	**0.112**	**0.181**	0.116	0.203	0.121	0.186	0.115	0.184	0.168	0.260	0.137	0.195	0.130	0.215	0.205	0.259	0.139	0.207	0.266	0.315	0.183	0.265	0.127	0.192
	Avg	**0.075**	**0.155**	0.087	0.173	0.096	0.172	0.092	0.172	0.127	0.227	0.092	0.166	0.107	0.195	0.140	0.219	0.088	0.172	0.208	0.274	0.127	0.223	0.094	0.171
PEMS03	12	**0.064**	**0.168**	0.066	0.170	0.066	0.169	0.075	0.182	0.087	0.195	0.069	0.174	0.068	0.172	0.092	0.210	0.076	0.186	0.105	0.219	0.084	0.197	0.081	0.191
	24	**0.085**	**0.193**	0.088	0.198	0.088	0.195	0.127	0.237	0.118	0.223	0.097	0.208	0.087	0.196	0.145	0.260	0.111	0.224	0.182	0.295	0.115	0.232	0.122	0.238
	48	**0.130**	**0.240**	0.166	0.277	0.132	0.242	0.213	0.319	0.174	0.267	0.164	0.275	0.132	0.246	0.261	0.354	0.190	0.290	0.317	0.410	0.166	0.285	0.197	0.309
	96	0.195	0.298	0.212	0.311	0.191	0.300	0.303	0.385	0.269	0.327	0.260	0.357	**0.186**	**0.285**	0.428	0.477	0.292	0.364	0.453	0.510	0.208	0.322	0.300	0.401
	Avg	**0.119**	**0.225**	0.133	0.239	**0.119**	0.227	0.180	0.281	0.162	0.253	0.148	0.254	**0.119**	**0.225**	0.232	0.325	0.167	0.266	0.264	0.359	0.143	0.259	0.175	0.285
PEMS08	12	**0.076**	**0.178**	**0.076**	**0.178**	0.078	0.179	0.087	0.191	0.092	0.199	0.088	0.193	0.133	0.184	0.101	0.216	0.090	0.199	0.112	0.222	0.110	0.223	0.094	0.200
	24	**0.105**	**0.209**	0.107	0.212	0.107	0.212	0.139	0.243	0.122	0.227	0.138	0.243	0.154	0.207	0.157	0.269	0.141	0.250	0.196	0.303	0.156	0.265	0.135	0.247
	48	**0.151**	**0.243**	0.164	0.255	0.158	0.248	0.259	0.345	0.174	0.269	0.338	0.358	0.216	0.258	0.296	0.371	0.242	0.323	0.385	0.424	0.232	0.323	0.233	0.330
	96	**0.230**	**0.306**	0.286	0.333	0.245	0.314	0.438	0.448	0.304	0.333	0.446	0.427	0.320	0.366	0.509	0.499	0.383	0.395	0.634	0.539	0.312	0.370	0.378	0.432
	Avg	**0.141**	**0.234**	0.158	0.245	0.147	0.238	0.231	0.307	0.173	0.257	0.253	0.305	0.206	0.254	0.266	0.339	0.214	0.292	0.332	0.372	0.203	0.295	0.210	0.302
NN5	24	**0.722**	**0.576**	0.743	0.582	0.775	0.601	0.780	0.608	0.764	0.615	0.785	0.607	0.822	0.646	0.743	0.581	0.853	0.663	0.805	0.631	0.877	0.668	0.836	0.644
	36	**0.685**	**0.569**	0.699	0.577	0.723	0.592	0.733	0.601	0.703	0.594	0.732	0.598	0.778	0.635	0.690	**0.569**	0.778	0.634	0.755	0.620	0.814	0.646	0.789	0.636
	48	**0.656**	**0.561**	0.680	0.576	0.699	0.588	0.719	0.604	0.677	0.585	0.717	0.601	0.805	0.657	0.710	0.584	0.758	0.629	0.742	0.622	0.818	0.653	0.785	0.642
	60	0.663	**0.572**	0.676	0.579	0.708	0.598	0.711	0.604	**0.650**	**0.572**	0.722	0.610	0.810	0.658	0.665	0.577	0.738	0.625	0.748	0.631	0.827	0.660	0.786	0.648
	Avg	**0.681**	**0.570**	0.699	0.578	0.726	0.595	0.736	0.604	0.698	0.592	0.739	0.604	0.804	0.649	0.702	0.578	0.782	0.638	0.763	0.626	0.834	0.657	0.799	0.643
1st Count		**26**	**30**	4	4	2	3	4	7	1	1	0	1	2	2	0	1	4	0	0	0	4	0	0	0

Table 5
The ablation experiment results of GPS-Mamba.

Data	Weather		ECL		Traffic		Solar		SD		PEMS03		PEMS08	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
w/o LTC	0.169	0.214	0.142	0.238	0.379	0.261	0.205	0.247	0.117	0.187	0.232	0.327	0.302	0.352
w/o STV	0.169	0.213	**0.138**	0.237	0.392	0.265	0.206	0.247	0.117	0.186	0.215	0.313	0.279	0.330
w/o LS	0.167	0.211	0.141	0.237	0.382	**0.260**	0.203	0.242	0.115	0.185	0.209	0.310	0.245	0.308
GPS-Mamba	**0.165**	**0.209**	0.140	**0.235**	**0.377**	**0.260**	**0.199**	**0.238**	**0.112**	**0.181**	**0.195**	**0.298**	**0.230**	**0.306**

- **w/o LTC**: We exclude the long-term correlation modeling module and replace it with simple skip connection.
- **w/o STV**: We remove the short-term dynamic variation modeling part and replace it with skip connection.
- **w/o LS**: The Laplace Scanning algorithm is not employed. It means that Mamba scans along the original permutation for spatial correlations modeling.

According to the experiment results illustrated in Table 5, the following conclusions can be drawn:

5.3.1. Effectiveness of LTC and STV

Different perspectives of multivariate correlations modeling both enhance the predictive accuracy of GPS-Mamba. Permutation equivariance is a major limitation for generalizability and robustness of multivariate correlations modeling. Especially for datasets with fine temporal granularity, the complex and dynamic spatial correlations further exacerbate the variation in retrieved scanning modes across different inputs. Modeling solely based on highly fluctuating scanning modes constrains the model's performance. Consequently, the fixed scanning mode in LTC enhances the convergence and stability of GPS-Mamba. As observed on the PEMS03 and PEMS08, their performance exhibits significant degradation in the absence of the LTC module. Meanwhile, the scanning mode of STV utilizes a retrieval mechanism to retrieve dynamic modes from the repository according to the inputs. It can capture short-term variation to obtain fine-grained data features, thereby improving prediction performance. Thus, combining both modules is crucial for GPS-Mamba.

5.3.2. Benefits of LS

Results of "**w/o LS**" across all datasets show a drop in performance. It demonstrates the effectiveness of the proposed Laplace Scanning algorithm. Specifically, the improvement of the algorithm on the ECL and Weather dataset is relatively slight, which may be due to the insignificant correlations between variables compared with other datasets. While in dataset SD, which contains plenty variables with irregular distribution and complex multivariate correlations, the algorithm significantly boost GPS-Mamba's performance. This result is consistent with the general perception that urban traffic network data have more intricate spatial connections. It suggests the importance of incorporating graph structure information and similarity between variables into sequence permutation for SSMs scanning.

5.4. Retrieval mechanism analysis

5.4.1. Data pattern analysis

As shown in the t-SNE visualizations of ECL and Traffic in Fig. 3, there are obvious differences between MTS across different periods. The phenomenon is consistent with cognition that MTS variations caused by human activities exhibit periodicity. At the same time, MTS within the same period also exhibit different patterns, which supports the rationale for obtaining representative samples from the training set through clustering algorithm for dynamic modeling.

5.4.2. Effect of retrieval repository

The operations of t-SNE embedding, DBSCAN, and cluster sampling during pre-processing are designed to extract a representative subset of MTS samples from the training set. The hyperparameter settings are configured based on the final sampled instance size n_s in the repository. For manual tuning across datasets, it is only necessary to adjust the neighborhood radius of DBSCAN according to n_s to control the number of

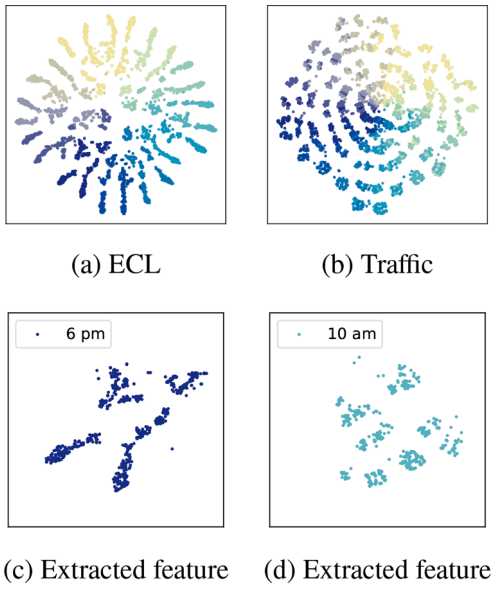


Fig. 3. Visualization results of t-SNE features. Features in different colors represent different time periods within a day. (c) and (d) show the extracted features from specific time periods in (a) and (b), respectively.

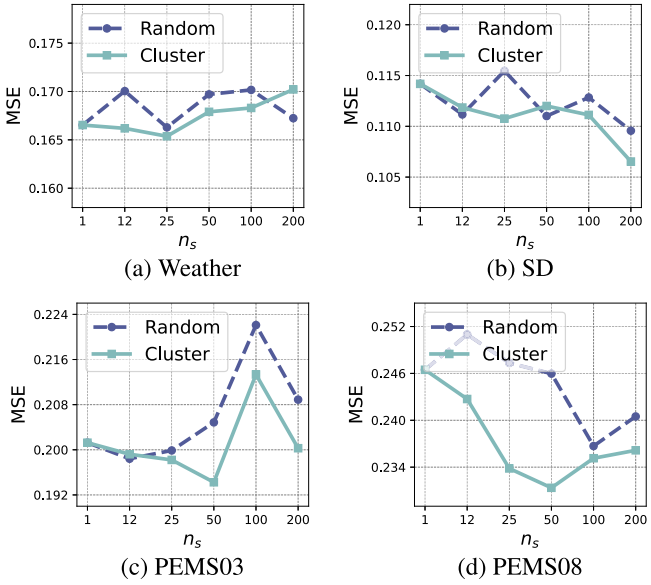


Fig. 4. Influence of cluster sample strategy and sample size in repository.

Table 6
Performance of different retrieval methods. "N/A" means no retrieval mechanism.

Data	ECL		Traffic		SD	
	MSE	MAE	MSE	MAE	MSE	MAE
N/A	0.152	0.246	0.384	0.268	0.111	0.178
Random	0.149	0.242	0.387	0.267	0.113	0.184
ED	0.142	0.237	0.384	0.263	0.110	0.179
PCC	0.142	0.237	0.382	0.262	0.107	0.178
DLinear	0.146	0.240	0.381	0.262	0.108	0.179
S-Mamba	0.140	0.235	0.378	0.260	0.106	0.177

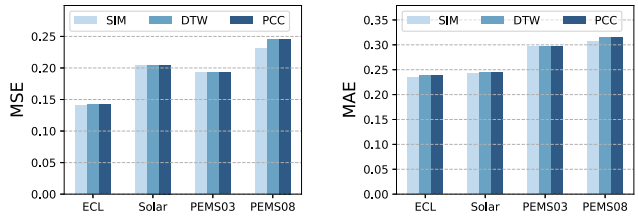


Fig. 5. Experiment results of MSE and MAE performance on constructing different similarity adjacency matrix.

clusters in various period instances c_i, without the need for excessive manual adjustments.

Here, we conduct experiment on the sample strategy and sample size n_s in retrieval process as presented in Fig. 4. Compared to the random sampling curve, the cluster sampling curve is generally lower under different circumstances. As cluster sampling is more beneficial for obtaining comprehensive and representative MTS samples, thereby providing more typical scanning modes from different patterns. As the number of samples n_s in repository increases within a certain range, the greater diversity of samples provides more valuable categorical scanning modes beneficial for prediction. However, excessively large sample size n_s can lead to suboptimality, possibly due to overfitting problem. In particular, the recommended sample sizes vary across datasets of different scales. Weather with a few variables and patterns performs best when n_s is around 25. Meanwhile, GPS-Mamba performs optimally when the sample size is set to 200 on SD which exhibits multiple patterns.

5.4.3. Impact of retrieval method

We conduct the experiments on different categories retrieval methods, including: (1) Remove retrieval mechanism as baseline. (2) Randomly retrieve scanning mode from the repository as baseline. (3) Retrieve similar reference directly by Euclidean distance (ED) or Pearson correlation coefficient (PCC). (4) Retrieve references through the embedding distance of pre-trained encoders DLinear or S-Mamba. The results are presented in Table 6.

Retrieval methods considering the correlation between MTS perform better than two naive baselines, demonstrating the effectiveness of retrieval mechanism. Meanwhile, embedding-based methods better reflect the similarity of MTS and slightly surpass correlation-based methods. However, the advantage comes at the cost of time and space overhead. Thus, embedding-based methods are preferable when resources are abundant, whereas correlation-based methods are suitable options under resource constraints.

5.5. Discussion on Laplace Scanning algorithm

5.5.1. Impact of adjacency matrix

In GPS-Mamba, we employ radial basis function kernel to measure the similarity between time series motivated by its efficiency and low computational complexity, especially in the case of large datasets or real-time applications. Since the scanning mode is derived from the Laplace Scanning algorithm applied to adjacency matrix, a consideration is whether the accuracy of similarity matrix would impact the results of model.

We conduct the experiments on constructing similarity matrix with radial basis function kernel (SIM) and two widely used indicators: Dynamic Time Warping (DTW) and Pearson correlation coefficient (PCC). As presented in Fig. 5, different similarity matrices yield slightly varying results, which aligns with the conclusion in Kljun et al. (2020). The selection of SIM in this work is primarily motivated by its efficiency and low computational complexity, which surpass DTW and PCC, especially when applied to large-scale datasets (Chen et al., 2024a).

We further explore the impact of learnable similarity function. Specifically, we incorporate time-varying vectors from TQNet (Lin et al., 2025) into GPS-Mamba to construct a learned inter-variable correlation matrix for the LS algorithm. As shown in Table 7, although the learnable matrix may capture finer-grained multivariate correlations, it does not lead to significant improvement in model performance and introduces additional computational overhead.

In certain datasets, both the graph structure and training data are available. We present their impact on GPS-Mamba in Table 8. GPS-Mamba generally exhibits superior performance when π_0 is derived from the physical graph structure. This may be attributed to the noise in datasets that can introduce spurious correlations, whereas the physical graph structure provides a more accurate representation of real-world multivariate relationships.

Table 7
Influence of learnable similarity matrix. "w/ TQ Matrix" means the similarity matrix in GPS-Mamba are constructed by the learned correlation matrix in TQNet.

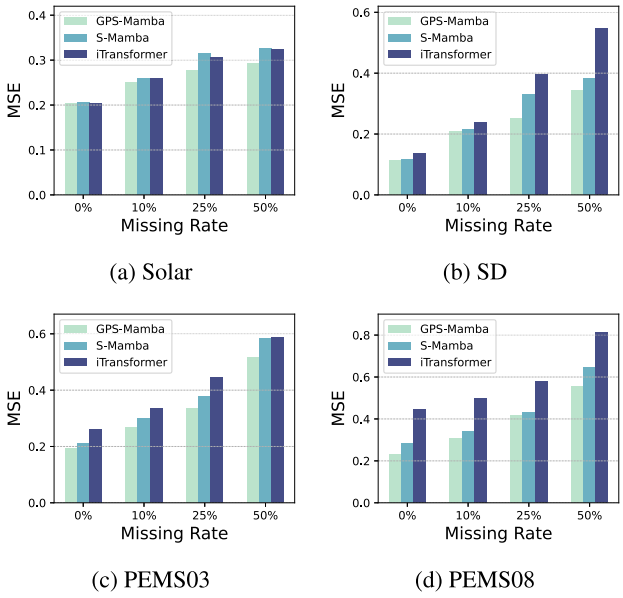
Data		ECL		Traffic	
Method	T	MSE	MAE	MSE	MAE
GPS-Mamba	96	0.140	0.235	0.379	0.260
	192	0.160	0.256	0.389	0.264
w/ TQ Matrix	96	0.140	0.236	0.381	0.262
	192	0.161	0.255	0.390	0.264

Table 8
Performance comparison between graph-structure-based π_0 and long-term dependency-based π_0.

Data	SD		PEMS03		PEMS08	
Metric	MSE	MAE	MSE	MAE	MSE	MAE
Graph	0.112	0.181	0.195	0.298	0.243	0.311
Dependency	0.124	0.186	0.201	0.301	0.243	0.312

5.5.2. Impact of structure information

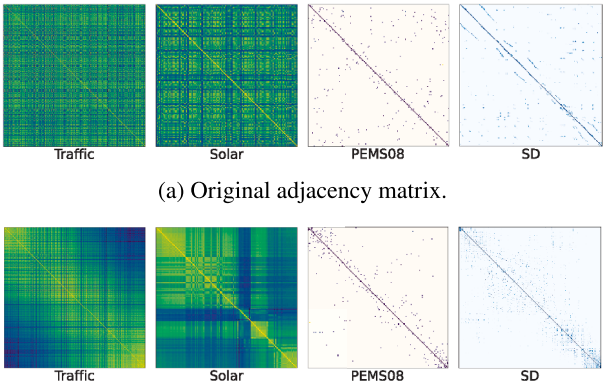
The goal of LS algorithm is incorporating graph or other implicit structure into sequence permutation. To verify the effectiveness of structure information in benefiting model to learn intrinsic patterns of variates, we compare GPS-Mamba with S-Mamba that utilizes the default variable permutation during Mamba modeling, and iTransformer with weak positional inductive bias. We train three models with the same missing variables with different proportions of missing variates. Subsequently, models are tasked with forecasting entire variates based on the learned information from the few variates. The outcomes of the investigation in Fig. 6 reveal that inherently permutation-invariant self-attention is incapable of extracting structure information and multivariate correlations among variables, thereby lacking generalization capability in different datasets. S-Mamba benefits from the sequential scanning paradigm of SSMs but leads to suboptimal performance due to naive scanning. In contrast, GPS-Mamba, enhanced by the proposed LS algorithm, accomplishes pattern transfer between relevant variables and demonstrate superior generalization potential.



(a) Solar (b) SD

(c) PEMS03 (d) PEMS08

Fig. 6. Generalization performance among GPS-Mamba, S-Mamba and iTransformer on different proportions of unseen variates.



(a) Original adjacency matrix.

(b) Adjacency matrix after applying LS algorithm.

Fig. 7. Visualize impact of LS algorithm in adjacency matrix.

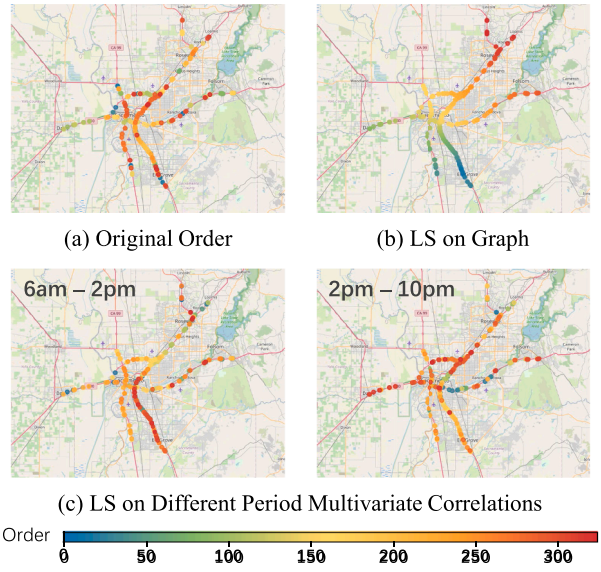
Table 9
Spearman's rank correlation coefficient between permutations before and after applying LS algorithm.

Weather	ECL	Traffic	Solar	SD	PEMS03	PEMS08	NN5
0.296	0.644	0.022	0.130	0.153	0.206	0.113	0.359

In contrast, GPS-Mamba benefits from the sequential scanning paradigm of SSMs and the proposed LS algorithm. It accomplishes pattern transfer between relevant variables and demonstrate superior generalization potential.

5.6. Empirical analysis

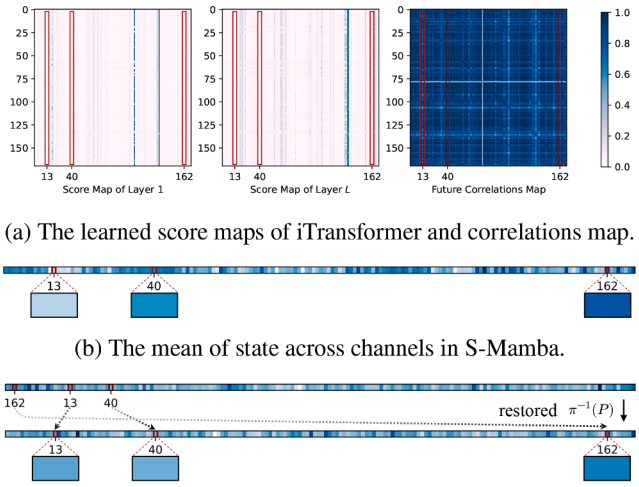
To verify the interpretability of GPS-Mamba, we visualize the changes applied by the LS algorithm to the similarity-based adjacency matrix for Traffic and Solar, and the distance-based adjacency matrix for SD and PEMS08. As shown in Fig. 7(a), the original adjacency matrix appears to be chaotic. While in adjacency matrix obtained from rearranging permutation with LS algorithm Fig. 7(b), the elements representing high correlations (yellow highlighted part in similarity-based adjacency matrix) or short distances (dark-colored parts in distance-based adjacency matrix) between variables are concentrated along the diagonal.



(a) Original Order (b) LS on Graph

(c) LS on Different Period Multivariate Correlations

Fig. 8. Order of road network sensors derived from specific rules.



(a) The learned score maps of iTransformer and correlations map.

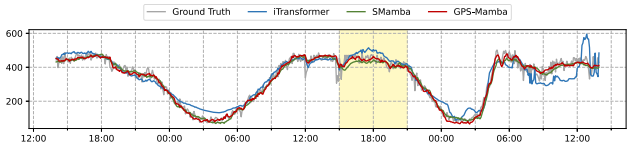
(b) The mean of state across channels in S-Mamba.

(c) The mean of state across channels in GPS-Mamba applying LS algorithm (above) and the restored original permutation (below).

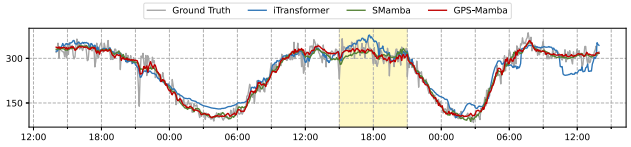
Fig. 9. A case visualization of the learned score maps by self-attention and learned state representation by SSMs.

This phenomenon implies that there is strong correlation between adjacent variables in the permutation, that is, the proximity of variables in high-dimensional space is maintained in sequence order. To systematically quantify the difference in permutations, we introduce Spearman's rank correlation coefficient (Sedgwick, 2014b) to measure the relative changes in rank order. The results are summarized in Table 9. Lower values indicates a greater difference between permutations generated by the LS algorithm and original permutations of datasets.

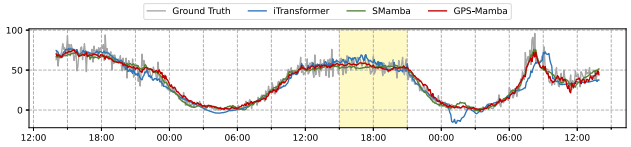
As shown in Fig. 8, we further visualize the permutation of variables on PEMS03 from the original order, to LS applied on the distance graph, and LS applied on the correlation matrix of different periods. Compared with the original order, the permutation incorporating graph structure better aligns with the real-world regional distribution. Thus, it can better capture the geographic semantic information between sensors. Permutation in Fig. 8(c) illustrate the dynamic nature of multivariate correlations over time, and reveal cross-regional similar patterns that are not identified in Fig. 8(a) and (b). This data-driven scanning modes enrich the representation of correlations between variables and reduces the



(a) Prediction curve for 576 steps across the 13th variable.



(b) Prediction curve for 576 steps across the 40th variable.



(c) Prediction curve for 576 steps across the 162th variable.

Fig. 10. Visualizing the forecast performance of different models on relevant variables.

noise caused by incomplete prior structure. Therefore, our GPS-Mamba can learn diverse relationships from distinct perspectives, which reflects real-world scenarios with favorable interpretability.

To gain an intuitive understanding of the superiority of GPS-Mamba over other baselines, we present the case visualization on series from PEMS08 in Fig. 9. Specifically, we highlight variables 13, 40 and 162 within the dataset. These variables exhibit high relevance and PCC manifested by their connectivity in real-world scenarios. It can be observed in Fig. 9(a) that both shallow attention layer and deep attention layer of iTransformer fail to comprehensively learn the multivariate correlations, evidenced by the disparity between the learned score maps and correlations matrix constituted by ground truth. As depicted in the output state of SSMs in Fig. 9(b), S-Mamba exhibits similar state for adjacent variables in permutation, while it portrays distinct representations for variables with long interval. It validates the positional inductive bias of SSMs. Despite the selection mechanism in Mamba, it struggles to retain crucial information during long scanning interval. In contrast, the intervals of relevant variables are significantly improved by the LS algorithm Fig. 9(c). The algorithm makes better use of the inductive biases of sequential paradigm, facilitating more effective semantic information propagation and noise filter within the model. Therefore, the output state representations of GPS-Mamba for variables 13, 40 and 162 exhibit greater similarity and closer approximation compared to S-Mamba.

We also visually compare forecasting curves between GPS-Mamba, S-Mamba, and iTransformer, as well as the ground truth curves, within specific time periods on variables 13, 40 and 162. As shown in Fig. 10, GPS-Mamba outperforms S-Mamba and iTransformer, particularly around period 18:00 (highlighted with yellow background). Specifically, our model can precisely forecast the consistent trend in time series caused by the collaborative changes among variables, thus approximating the ground truth. Other models lacking such ability may result in incorrect pattern change prediction and performance degradation.

5.7. Hyper-parameter study

5.7.1. Model pipeline

Fig. 11 draws the impact of hyper-parameters on the Weather, ECL. Traffic and Solar datasets. We search the number of encoder layers l_1 in LTC and l_2 in STV from a search space of $[1, 2, 3, 4]$. Moreover, we search the number of dimension d from a search space of $[128, 256, 512, 1024]$.

Table 10
Comprehensive performance analysis with training time (s) and inference time (s).

Model	ECL		Traffic		SD	
	Train	Inf.	Train	Inf.	Train	Inf.
PatchTST	276.19	36.09	345.75	36.83	736.69	85.02
Crossformer	245.05	33.15	951.96	99.10	1407.82	169.22
iTransformer	58.12	116.49	104.33	95.78	141.54	200.26
KARMA	77.52	144.33	155.72	125.35	163.59	222.53
S-Mamba	64.31	66.25	117.18	71.14	240.79	244.19
SDE-Mamba	756.48	161.54	558.82	135.93	806.52	284.17
GPS-Mamba	98.51	90.56	143.34	78.12	230.74	198.62

First, the performance of GPS-Mamba improves as the layers of the encoder increase and tends to be best when l_1 reaches 4. This can be attributed to the deeper layer structure, which is conducive to the modeling of complex temporal dependencies and multivariate correlations. Second, when the number of l_2 reaches a certain threshold, the performance of GPS-Mamba remains relatively stable and may even show a decline. The threshold is 3 in most of our experimental settings. Third, our model achieves optimal performance when the parameter d is set to 512. A small dimension size makes the model's performance suffer due to the acquisition of sufficient representation within finite space, and a large dimension size may result in overfitting problem.

5.7.2. LS algorithm

We investigate the impact of scaling factors θ, σ, and threshold ϵ in Laplace Scanning algorithm. As shown in Fig. 12, escalating the scaling factor progressively reduces the disparities among variables. The attenuation of extreme-value may yield permutation that is more conducive for GPS-Mamba. However, an excessive number of scaling factor would over-smooth correlation differences, ultimately impairing model performance. The threshold ϵ can filter out weak inter-variable correlations and noise, while simultaneously risking the exclusion of informative signals. The optimal values vary depending on the specific dataset.

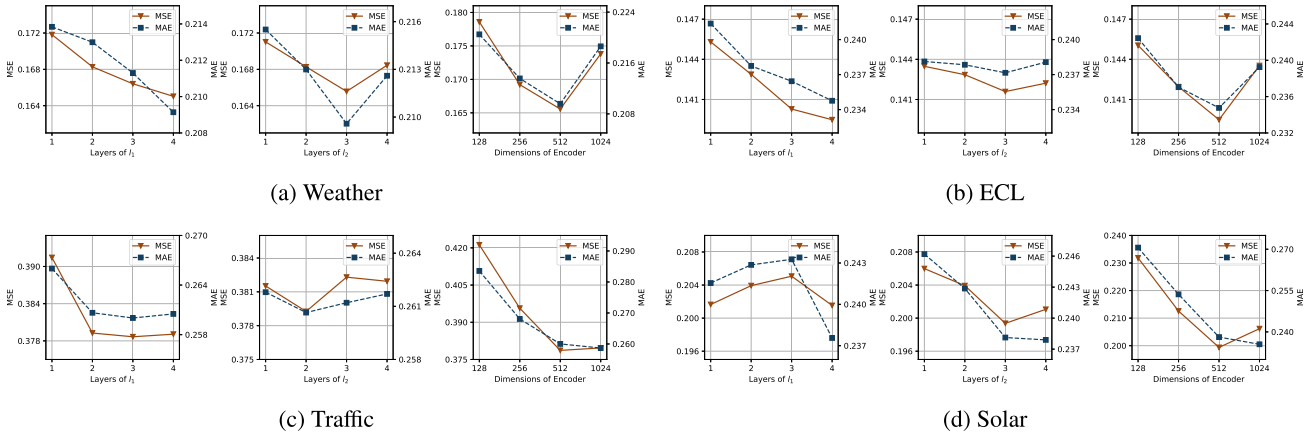
5.7.3. Selective SSM architecture

As presented in Fig. 13, the number of latent state dimension N in GPS-Mamba had variation from 2 to 32 to facilitate the retention of richer historical information and more comprehensive multivariate dependencies. Nevertheless, the performance degrades when N becomes excessively large, possibly due to overfitting noise in the data.

5.8. Model efficiency analysis

In terms of computational complexity, our proposed GPS-Mamba model primarily involves the pre-processing phase, training and inference phases. During the pre-processing stage, the complexity of the pre-trained encoder depends on the specific encoders selected. For π_0 constructed from long-term multivariate correlations of training data, the computational cost varies depends on both the length MTS and the number of variables. For scanning mode π_X built from fixed-length MTS ($L = 96$), the overall cost depends on both the number of variables and the number of representative samples n_s in the repository. Fig. 14 presents the computational cost of the LS algorithm under the relevant settings. The computational resources in pre-training phase can be regarded as additional overhead. Here, we primarily focus on comparing the computational cost of GPS-Mamba with the benchmark models during both the training and inference stages.

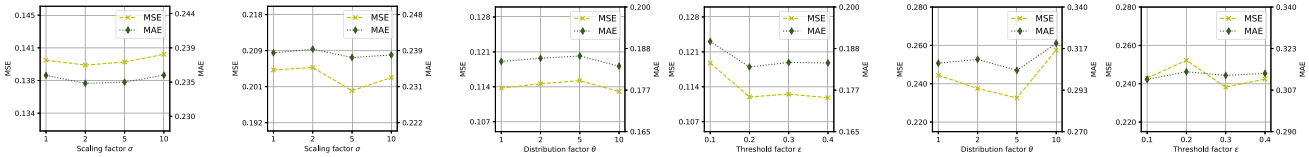
Results in Table 10 indicate that GPS-Mamba does not exhibit a significant efficiency advantage, which can be attributed to the reasons as follows. (1) Although Mamba has been demonstrated to exhibit linear complexity in many studies (Gu & Dao, 2023), the relatively complex basic structure of it differs from Transformer that has been well optimized on hardware (Qu et al., 2024). Consequently, the Mamba-based



(a) Weather (b) ECL

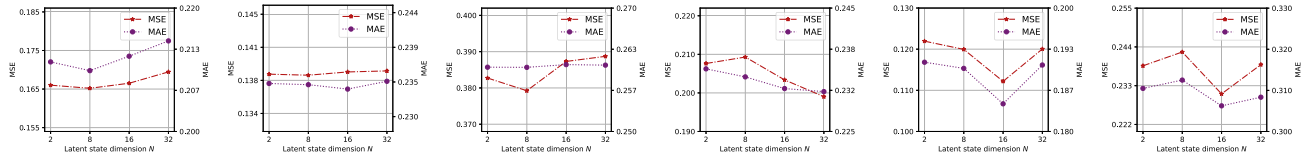
(c) Traffic (d) Solar

Fig. 11. Hyperparameter study on the number of layers l_1, l_2 and dimension d.



(a) ECL (b) Solar (c) SD (d) PEMS08

Fig. 12. Hyperparameter study on scaling factors θ and σ, and threshold factor ϵ.



(a) Weather (b) ECL (c) Traffic (d) Solar (e) SD (f) PEMS08

Fig. 13. Hyperparameter study on the latent state dimension N of Selective SSM.

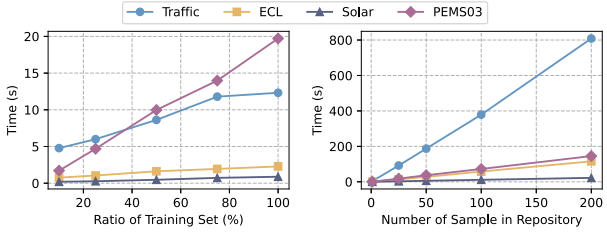


Fig. 14. Computational cost of the LS algorithm in the pre-processing stage.

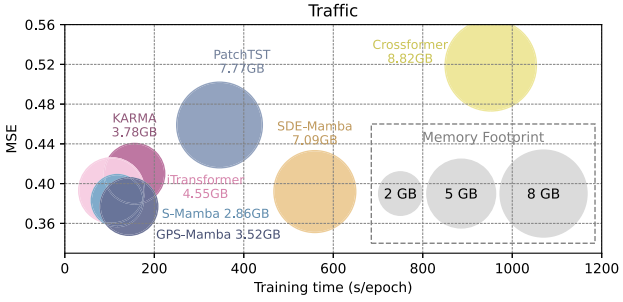


Fig. 15. Model efficiency and performance comparison.

architecture does not significantly save computational resources on ECL with fewer variables compared to Traffic and SD. It implies that our model is more applicable in scenarios involving large-scale variables. (2) Our model employs a retrieval mechanism, which incurs additional I/O operations cost that cannot be efficiently parallelized on GPU. It may be further optimized in future explorations. (3) Models that consider inter-variable correlations exhibit more pronounced variations in computational cost as the number of variables increases. This observation stems from our experimental settings where input and output lengths are constant. In contrast, models focusing solely on temporal dependencies demonstrate smaller fluctuations in time overhead, albeit at the cost of compromised predictive performance. Overall, as shown in Fig. 15, considering the balance between prediction capability and computational cost across all models, the computation cost of our GPS-Mamba is acceptable.

5.9. Discussion

In this work, we utilize the inductive bias of Mamba to precisely model complex dependencies among relevant variables while simultaneously filtering out irrelevant noise. The proposed Laplace Scanning algorithm better aligns with the inherent nature of MTS compared with naive scanning mode. Moreover, the encoder layer thoroughly extracts MTS patterns by integrating long-term stable correlations and short-term dynamic variation from distinct views. The experimental results

in Table 4 demonstrate that our model consistently achieves either the best or second-best predictive performance across different settings. This provides strong evidence of its broad applicability from the perspective of predictive capability.

However, since GPS-Mamba primarily focuses on leveraging Mamba to model comprehensive multivariate correlations among variables with effective permutation, its improvements over baseline models are more pronounced on datasets exhibiting stronger and more complex dynamic spatial dependencies (e.g., PEMS08) than on datasets with weaker inter-variable correlations (e.g., Weather). Additionally, GPS-Mamba employs a retrieval mechanism that incurs extra CPU and I/O overhead, which constrains the computational efficiency. As a result, our method does not exhibit a computational advantage over the baseline models on certain small-scale datasets. The optimization of retrieval mechanism remains a challenging and worthwhile direction for future work. Compared to existing methods, GPS-Mamba is preferable for high-dimensional datasets exhibiting complex spatial dependencies (e.g., energy and traffic scenarios). It is particularly suited for applications that prioritize predictive accuracy over computational efficiency, thereby justifying a slight trade-off in efficiency for improved performance.

6. Conclusion

In this paper, we propose an effective Mamba framework called GPS-Mamba (Graph Permutation Scanning State Space Model) for multivariate time series forecasting. We propose a novel Laplace Scanning algorithm, which incorporates graph structure information and multivariate correlations into sequence permutation to enhance SSMs' ability of extracting correlation. Then, we utilize Mamba block with the proposed algorithm to encode multivariate correlations from different views. In long-term correlation modeling, the fixed scanning mode is based on the stable feature of datasets. While in short-term variation modeling, retrieval mechanism is employed to dynamically match representative scanning modes within the preprocessed repository. Finally, we fuse features obtained from two perspectives to precisely capture multivariate correlations. Extensive experimental results show that GPS-Mamba outperforms a range of existing models across seven real-world datasets with high interpretability. In future research, we aim to explore more efficient retrieval methods to improve the generalization and applicability of model in different scenarios.

CRediT authorship contribution statement

Zihao Yao: Conceptualization, Formal analysis, Methodology, Validation, Visualization, Writing – original draft; **Qi Zheng:** Methodology, Validation, Writing – original draft, Writing – review & editing; **Jiankai Zuo:** Validation, Investigation, Formal analysis; **Yaying Zhang:** Investigation, Funding acquisition, Project administration, Writing - review & editing.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

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